🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.DS.5

Represent and Describe Data in a Box Plot

Lesson 2-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can make and read a box plot to summarize a data set.
Language Objective: I can explain my box plot using the words box plot, median, quartile, and interquartile range.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Display Data with Box Plots | Box Plot | Median | Quartile | Interquartile Range | Data distribution | Variability

👆 *Tap any word to see what it means and a picture.*

1. ________ — Show a data set's five-number summary in a box-and-whisker display.

2. A graph that shows data with a box and whiskers based on the five-number summary is a ________.

3. The middle value of the data, shown as a line inside the box, is the ________.

4. A value that divides the data into four equal parts is a ________.

5. The distance between the first and third quartiles, showing the middle 50% of data, is the ________.

6. The way data values are spread out is the ______________________.

7. How spread out the data values are is the ______________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Which team is more consistently scoring high? Use the box plot values to support your answer?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Display Data with Box Plots** — Show a data set's five-number summary in a box-and-whisker display.
  Representar datos con diagramas de caja — Mostrar el resumen de cinco números de un conjunto de datos en un diagrama de caja y bigotes.
- [ ] **Box Plot** — A graph that shows how data is spread using five key numbers.
  Diagrama de caja — Una gráfica que muestra cómo se reparten los datos con cinco números clave.
- [ ] **Median** — The middle number when you put them in order.
  Mediana — El número del medio cuando los pones en orden.
- [ ] **Quartile** — Numbers that split the data into four equal parts.
  Cuartil — Números que dividen los datos en cuatro partes iguales.
- [ ] **Interquartile Range** — The spread of the middle half of the data (Q3 – Q1).
  Rango intercuartílico — La extensión de la mitad central de los datos (Q3 – Q1).

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

Team A's IQR is ____ and Team B's IQR is ____. Team B's middle 50% scores between ____ and ____. The fan is wrong because ____. The box plot shows that Team B is more consistent because its Q1 and median are higher and its IQR (10) is smaller, while ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A box plot is built from the minimum, Q1, median, Q3, and maximum.

Evidence: Students name the five-number summary (min, Q1, median, Q3, max) and find the median (14) for the 11 ordered scores.

Reasoning: This evidence connects the definition of box plot to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Team A's IQR is ____ and Team B's IQR is ____. Team B's middle 50% scores between ____ and ____. The fan is wrong because ____. The box plot shows that Team B is more consistent because its Q1 and median are higher and its IQR (10) is smaller, while ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Display Data with Box Plots
2. **Notes 2:** Box Plot
3. **Notes 3:** Median
4. **Notes 4:** Quartile
5. **Notes 5:** Interquartile Range
6. **Notes 6:** Data distribution
7. **Notes 7:** Variability
8. **Watch for this mistake:** *A common mistake in Display Data: Box Plots is treating a quartile as just one data point instead of the median of a half — for the lower half 4, 8, 10, 14, some students pick 10 (the last value) as Q1 instead of finding the median of that half: Q1 = (8 + 10) ÷ 2 = 9. A second common slip is finding IQR by adding the quartiles instead of subtracting, writing IQR = Q3 + Q1 (like 30 + 18 = 48) instead of IQR = Q3 − Q1 (30 − 18 = 12) — always order the data first, then subtract Q1 from Q3 to find the spread of the middle 50%.*
9. **Try It 1:** A. Q1 = 12, Q3 = 28 — *With 8 values the median splits the data into a lower half (6, 10, 14, 18) and an upper half (22, 26, 30, 34). Q1 is the median of the lower half: (10 + 14) ÷ 2 = 12. Q3 is the median of the upper half: (26 + 30) ÷ 2 = 28.*
10. **Try It 2:** A. Between 15 and 30 — *The box of a box plot stretches from Q1 to Q3, and that box always holds the middle 50% of the data. Here the box goes from Q1 = 15 to Q3 = 30.*